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Domain :- AI - ML

Branch Name :- Computer Engineering

1)

+ Code + Text

```
df.head(10)
```

	Car ID	Brand	Year	Engine Size	Fuel Type	Transmission	Mileage	Condition	Price	Model
0	1.0	Tesla	2016.0	2.3	Petrol	Manual	114832.0	New	26613.92	Model X
1	2.0	BMW	2018.0	4.4	Electric	Manual	143190.0	Used	14679.61	5 Series
2	3.0	Audi	2013.0	4.5	Electric	Manual	181601.0	New	44402.61	A4
3	4.0	Tesla	2011.0	4.1	Diesel	Automatic	68682.0	New	86374.33	Model Y
4	5.0	Ford	2009.0	2.6	Diesel	Manual	223009.0	Like New	73577.10	Mustang
5	6.0	Audi	2019.0	2.4	Diesel	Automatic	246553.0	Like New	88969.76	Q7
6	7.0	Audi	2020.0	4.0	Electric	Automatic	135486.0	Used	63498.75	Q5
7	8.0	Tesla	2017.0	5.3	Hybrid	Automatic	83030.0	New	17381.19	Model Y
8	9.0	Honda	2023.0	5.7	Electric	Manual	120360.0	Like New	15905.62	Civic
9	10.0	Ford	2010.0	1.5	Electric	Automatic	135009.0	Like New	9560.22	Explorer

Next steps: Generate code with df New interactive sheet

It contains features like **brand, model, year, mileage, and fuel type** that affect a car's price. The target variable is **Price**, which is a continuous numeric value.

2)

```
df.isnull().sum()
```

	0
Car ID	0
Brand	0
Year	0
Engine Size	0
Fuel Type	0
Transmission	0
Mileage	0
Condition	0
Price	0
Model	0

dtype: int64

3)

```
df.isnull().sum() / len(df) * 100
```

	0
Car ID	0.0
Brand	0.0
Year	0.0
Engine Size	0.0
Fuel Type	0.0
Transmission	0.0
Mileage	0.0
Condition	0.0
Price	0.0
Model	0.0

dtype: float64

4)

```
df.select_dtypes(include=['number']).columns
```

```
Index(['Car ID', 'Year', 'Engine Size', 'Mileage', 'Price'], dtype='object')
```

5)

```
df[numeric_cols].fillna(df[numeric_cols].median())
```

	Car ID	Year	Engine Size	Mileage	Price
0	1.0	2016.0	2.30	114832.0	26613.92
1	2.0	2018.0	4.40	143190.0	14679.61
2	3.0	2013.0	4.50	181601.0	44402.61
3	4.0	2011.0	4.10	68682.0	86374.33
4	5.0	2009.0	2.60	223009.0	73577.10
...	...	...	...	...	...
2495	2496.0	2020.0	2.40	22650.0	61384.10
2496	2497.0	2001.0	5.70	77701.0	24710.35
2497	1249.5	2012.0	3.45	149762.0	53485.24
2498	2499.0	2002.0	4.50	229164.0	46085.67
2499	2500.0	2005.0	4.60	80978.0	16594.14

2500 rows × 5 columns

6)

```
df["Price"].min()
```

```
5011.27
```

```
df["Price"].max()
```

```
99982.59
```

```
df["Price"].mean()
```

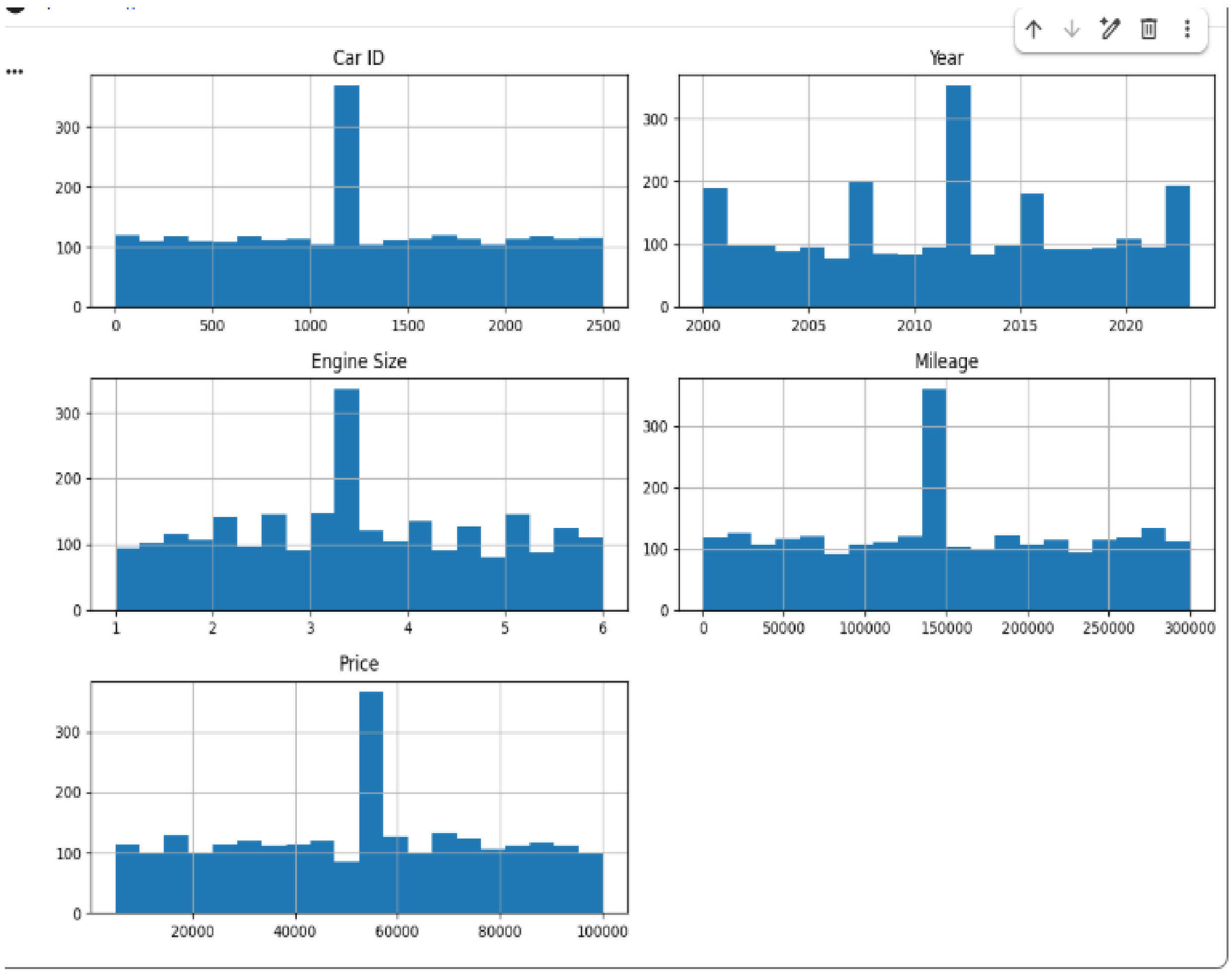
```
np.float64(52604.710952)
```

```
df["Price"].median()
```

```
53485.240000000005
```

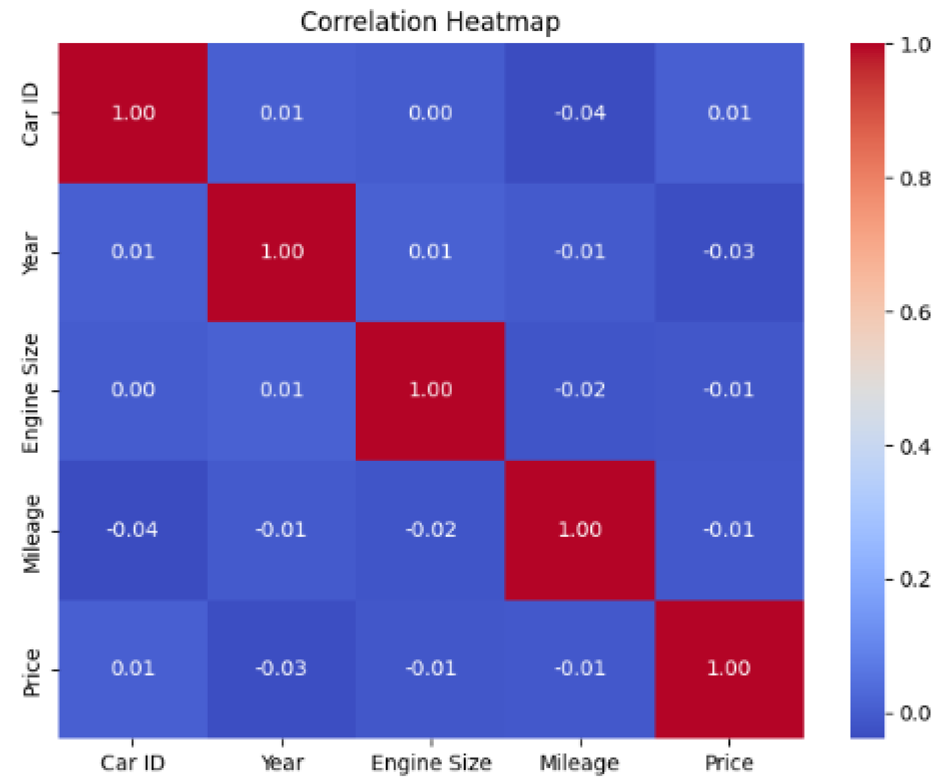
The **Price** column has a wide range of values with some variation. The difference between the **mean** and **median** suggests a slightly skewed distribution.

7)



Most cars are newer, have moderate mileage, and fall within a mid-range price. A few extreme values (outliers) are present, especially in the **Price** column, making the data slightly skewed.

8)



9)

```
X = df.drop("Price", axis=1)
y = df["Price"]
X.head()
y.head()
```

	Price
0	26613.92
1	14679.61
2	44402.61
3	86374.33
4	73577.10

dtype: float64

10)

	Car ID	Year	Engine Size	Mileage	Brand_BMW	Brand_Ford	Brand_Honda	Brand_Mercedes	Brand_Tesla	Brand_Toyota	...	M
0	-1.816308	0.661432	-0.872803	-0.422884	-0.387239	-0.374154	-0.37137	-0.390647	2.638519	-0.559487	...	
1	-1.814855	0.963456	0.677967	-0.083709	2.582385	-0.374154	-0.37137	-0.390647	-0.379000	-0.559487	...	
2	-1.813402	0.208396	0.751813	0.375706	-0.387239	-0.374154	-0.37137	-0.390647	-0.379000	-0.559487	...	
3	-1.811949	-0.093627	0.456428	-0.974860	-0.387239	-0.374154	-0.37137	-0.390647	2.638519	-0.559487	...	
4	-1.810496	-0.395651	-0.651264	0.870965	-0.387239	2.672699	-0.37137	-0.390647	-0.379000	-0.559487	...	

5 rows × 43 columns

11)

	Brand	Fuel Type
0	Tesla	Petrol
1	BMW	Electric
2	Audi	Electric
3	Tesla	Diesel
4	Ford	Diesel

The Car Price Prediction Dataset contains information about different cars, including their brand, model, year, mileage, engine size, fuel type, and transmission. The objective is to predict the **Price** of a car, which is a continuous target variable. The dataset was cleaned by handling missing values and removing duplicates.